Adjoint Sensitivities and Gradient-Based Manning Roughness Calibration in RDycore

DRAFT for internal review

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Abstract

We present a discrete-adjoint capability for RDycore, the exascale river dynamical core of the Energy Exascale Earth System Model (E3SM), and use it for gradient-based calibration of spatially distributed Manning roughness. The capability has three parts. First, an exact assembled Jacobian of the finite-volume shallow-water right-hand side: closed-form Roe-flux blocks obtained by hand-written forward-mode differentiation of the flux computation, entropy-fix branches included, together with source-term and boundary-condition blocks. Second, adjoint sensitivities of gauge-misfit functionals with respect to both the initial state and the Manning field via PETSc's `TSAdjoint`; one gradient costs roughly one extra forward solve, independent of the parameter count. Third, bound-constrained calibration with TAO. Every derivative is validated against finite differences of the full nonlinear solve at tight, pre-registered tolerances (Jacobian blocks to 10^{-8}, gradients to 10^{-5}). Twin experiments recover a two-zone roughness field to 4×10^{-7} relative error, and a per-cell field to 8% with multi-time observations and Tikhonov regularization. The same assembled Jacobian enables fully implicit backward-Euler stepping, and we show implicit stepping is required in practice: explicit Manning drag is unstable at the operational time step of a Hurricane Harvey test case, in quantitative agreement with a friction-stiffness stability analysis. The implicit integrator's adjoint passes the same finite-difference gates. On a Hurricane Harvey twin configuration (2,746 cells, implicit stepping) the calibrated Manning map recovers a two-zone truth field to 19.4% with proper Tikhonov regularization. The gauge-sparse setting exhibits — and lets us dissect — the classic semiconvergence of underdetermined friction estimation: a twin using the actual 17-gauge USGS network geometry fits its 238 observations essentially perfectly while recovering the field to only 45%, quantifying why real-data calibration must shrink the parameter space to match the network — and, re-asking identifiability at basin scale, that the observable matters as much as the parameterization: peak water-surface elevations at 108 surveyed high-water-mark locations identify the land-cover roughness classes covering 83% of a 2.93M-cell Hurricane Harvey domain, where 13 stream gauges identify almost none. Within E3SM, adjoint-based parameter optimization is established in the land-ice component MALI, whose velocity solver inverts for basal friction by automatic differentiation of the *steady* momentum balance; RDycore's capability is, to our

knowledge, the first *transient* discrete adjoint in an E3SM component — differentiated through the time integrator itself — and it is obtained without rewriting the solver in a machine-learning framework.

1 Introduction

Manning roughness is the dominant tunable parameter of depth-averaged flood models, and its calibration is traditionally manual or ensemble based. Adjoint methods deliver the gradient of an observation misfit with respect to *every* cell's roughness at the cost of about one extra forward solve, making distributed calibration tractable. The approach is well established in shallow-water modeling, from channel networks [1] and unstructured finite-volume twin experiments [2] to GPU-accelerated optimizers [3]. It is deployed in variational-assimilation platforms such as DassFlow [4, 5] and the Thetis coastal model, which calibrated a spatially varying Manning field for the Baltic Sea with an automatically generated adjoint [6], following storm-surge sensitivity studies that addressed the overfitting inherent in distributed friction estimation [7].

A recent generation of *differentiable* flood and hydrology codes obtains gradients by rewriting simplified solvers inside machine-learning frameworks: Inunda (PyTorch, local-inertial, calibrated on a Hurricane Harvey hindcast) [8], Hydrograd (Julia) [9], AegirJAX [10], CaMa-Flood-GPU [11], and differentiable routing [12]. These demonstrate demand for exactly this capability, but none is a production Earth-system component: they trade the validated numerics, unstructured meshes, and coupling infrastructure of codes like RDycore for differentiability.

This paper takes the complementary route: make the production code differentiable. The route has precedent inside E3SM itself: the land-ice component MALI [13] routinely inverts for basal friction with an adjoint obtained by automatic differentiation of its Albany velocity solver [14] — but that is the adjoint of a *steady* nonlinear solve, used for initialization; no E3SM component has, to our knowledge, a discrete adjoint of its transient time stepper. RDycore is a finite-volume shallow-water dynamical core (Roe fluxes, well balancing, wetting and drying) validated at up to 471M cells. We add the missing derivative infrastructure — an exact assembled Jacobian and TSAdjoint-based sensitivities [15] — entirely behind a configuration flag, with no impact on existing runs and no new dependencies.

Contributions: (i) an exact, FD-verified Jacobian of the SWE right-hand side, including Roe dissipation, entropy-fix branches, wet/dry branches, and reflecting/Dirichlet/critical-outflow boundary conditions; (ii) adjoint gradients dJ/du_0 and dJ/dn with respect to water-height observations at gauge locations at multiple observation times, FD-validated to 10^{-5} or better; (iii) TAO-based recovery of per-region and per-cell roughness fields in twin experiments, with a quantified observability study; (iv) implicit (backward Euler) stepping and its adjoint, enabled by the same Jacobian, with an empirical demonstration that explicit friction fails at operational time steps on a Hurricane Harvey configuration; and (v) a catalog of the nondifferentiable structures encountered — useful, we believe, to anyone retrofitting adjoints onto an operational flood code.

2 Forward model

RDycore solves the 2D shallow water equations in conservative variables $u = (h, q_x, q_y)$, $\mathbf{q} = h\mathbf{v}$:

$$\partial_t h + \nabla\cdot\mathbf{q} = Q_r(x,t), \tag{1}$$

$$\partial_t \mathbf{q} + \nabla\cdot\left(\frac{\mathbf{q}\otimes\mathbf{q}}{h} + \frac{gh^2}{2}\,\mathbb{I}\right) = -\,gh\nabla z_b \;-\; g\,n^2 h^{-7/3}\,\mathbf{q}\,\|\mathbf{q}\|, \tag{2}$$

with cell-centered finite volumes on an unstructured mesh:

$$\frac{du_i}{dt} = -\frac{1}{|\Omega_i|} \sum_{e \in \partial\Omega_i} \mathcal{F}\big(u_i, u_j, \mathbf{n}_e\big)\, L_e + S(u_i; n_i, t) \equiv f_i(u, n), \tag{3}$$

where $\mathcal{F}$ is the Roe flux with a critical-flow (entropy) fix, primitives are reconstructed with a regularized division and a dry cutoff $h_{\min}$, and S contains bed slope, Manning drag, and external forcing.

A differentiable right-hand side. A prerequisite that deserves emphasis: RDycore's existing friction treatments (a semi-implicit splitting and the implicit scheme of Xia and Liang) embed the time step — and, in the semi-implicit case, the flux divergence — *inside* the RHS callback. The resulting map is consistent only with forward Euler and admits no well-defined $\partial f/\partial u$. We therefore added a plain explicit source treatment (`source.method: explicit`), used on every adjoint path; Section 6 shows how the resulting stiffness is then handled properly, by the integrator rather than inside f.

3 Exact Jacobian

The RHS Jacobian is block sparse with 3×3 blocks on the cell-adjacency graph: each interior edge (i, j) contributes $\mp(L_e/|\Omega_{i,j}|)\, \partial\mathcal{F}/\partial u_{L,R}$ to the two cell rows, local sources contribute diagonal blocks, and boundary edges contribute diagonal blocks through the ghost-state map of the boundary condition.

Flux blocks by forward-mode differentiation. Rather than deriving the dissipation-matrix derivatives symbolically, we differentiate the flux *computation*: a hand-written forward-mode (tangent) version of the Roe flux routine propagates directionals through every intermediate — Roe averages, eigenvectors, eigenvalues including both branches of the critical-flow fix, wave strengths, and physical fluxes — taking the code's own branch at every $|\cdot|$ and switch (a consistent one-sided derivative). Seeding unit directions through the primitive-reconstruction chain rule yields the two conservative-space blocks in six tangent evaluations per edge. The approach mirrors the flux code line by line, which makes it reviewable and, in our experience, correct on first assembly against the verification harness below.

Boundary blocks. For a boundary edge, $F(u_{\text{in}}, u_g(u_{\text{in}}))$ contributes $(\partial_L F + \partial_R F\, \partial u_g/\partial u_{\text{in}})$. The reflecting map is a linear mirror; Dirichlet ghosts are constant; critical outflow has a closed-form ghost map $h_g = (q^2/g)^{1/3}$ with the code's inflow branch producing an identically zero flux.

Parameter Jacobian. Manning n appears only in the drag term, so $\partial S_{\mathbf{q}}/\partial n = -2gnh^{-7/3}\mathbf{q}\|\mathbf{q}\|$: two nonzeros per cell, supplied to `TSSetRHSJacobianP`.

Verification. Three layers, all against central finite differences with pre-registered tolerances (Table 1): an edge-level harness that differences the flux routine itself on sampled states (with a Richardson self-check of the FD order); full-matrix comparisons of the assembled Jacobian against a matrix-coloring FD Jacobian of the actual RHS on three state/boundary combinations, boundary rows included; and end-to-end gradient checks through the adjoint (next section). A finite-difference coloring Jacobian (`numerics.jacobian: fd`) remains available permanently as the verification baseline.

Check	Configuration	Result	Gate
Edge flux blocks	wet jump 10 m/5 m	7.6×10^{-9}	10^{-6}
Edge flux blocks	transcritical (entropy-fix branch)	2.4×10^{-10}	10^{-5}
Source block	shallow flowing state	$< 10^{-6}$	10^{-6}
Assembled matrix	uniform flow, reflecting BCs	1.6×10^{-8}	10^{-6}
Assembled matrix	dam break, reflecting BCs	1.5×10^{-8}	10^{-6}
Assembled matrix	dam break, Dirichlet + critical outflow	1.9×10^{-8}	10^{-6}
dJ/du_0 (explicit RK)	8 sampled components	1.0×10^{-8}	10^{-5}
dJ/dn (explicit RK)	domain aggregate	2.9×10^{-6}	10^{-5}
dJ/du_0 (implicit BE)	tight inner tolerances	1.4×10^{-8}	10^{-5}
dJ/dn (implicit BE)	tight inner tolerances	5.3×10^{-6}	10^{-5}

Table 1: Relative errors of analytic derivatives against central finite differences of the full nonlinear computation. All checks run in CI.

4 Adjoint sensitivities

For gauge observations y_k at times t_k with observation operator H (one row per gauge, selecting water height) and error variance σ^2,

$$J(n) = \tfrac{1}{2\sigma^2} \sum_k \|Hu(t_k) - y_k\|^2. \tag{4}$$

`TSAdjoint` [15] differentiates the discrete time stepper, so gradients are exact to roundoff for the computed trajectory — the property that makes the FD gates in Table 1 meaningful. Multi-time observations are handled by a segmented backward sweep: the adjoint is advanced between observation times with `TSAdjointSetSteps` and receives a jump $H^T \sigma^{-2} r_k$ at each t_k (Algorithm 1 in Section 5). One sweep yields both dJ/du_0 (state sensitivity) and dJ/dn (per-cell parameter gradient, via the two-nonzero parameter Jacobian).

Two integrator facts constrain the design. PETSc's plain forward-Euler integrator implements no adjoint, so adjoint-path configurations use the Runge–Kutta family. And for implicit integrators the discrete-adjoint gradient is only as exact as the inner solves: with default SNES/KSP tolerances the Manning gradient checked at 2×10^{-4}; tightening the tolerances recovered 5×10^{-6} (Table 1). We state the production side of this plainly, because the verification numbers are otherwise easy to over-read: at the loose inner tolerance used for the large implicit runs (Krylov relative tolerance 10^{-2}, chosen because it is $\sim$2.5× cheaper end to end), the gradient is inner-solve-limited at 6×10^{-4}–4×10^{-3} relative. In every calibration we have run, the loose- and tight-tolerance settings converge indistinguishably (same objective to 5–6 digits), consistent with the tolerance of quasi-Newton methods to bounded gradient error; our policy is to run campaigns loose and repeat one calibration at Krylov tolerance 10^{-3} — where the 10^{-5} gates pass — as a spot check.

5 Calibration twin experiments

The complete calibration loop is summarized in Algorithms 1–3. It is a strong-constraint 4D-Var iteration: the model is enforced exactly (the only controls are the Manning field and, optionally, the initial state). Each objective evaluation performs one checkpointed forward sweep that records the gauge residuals r_k at the observation times, followed by one backward sweep of the discrete adjoint (Algorithm 2). The per-iteration cost is therefore one forward plus one adjoint solve, independent of the number of parameters.

The backward sweep carries two vectors with distinct roles. The *adjoint state* λ holds $\partial J_{\text{mis}}/\partial u(t)$ at the current point of the backward traversal. It is seeded with the terminal observation sensitivity $\sigma^{-2}H^T r_K$, propagated one step at a time by the transposed Jacobian of the forward step map (the backward "model step"), and incremented by a jump $\sigma^{-2}H^T r_k$ each time the sweep crosses an observation time, because the state at that time enters the misfit directly. The *parameter sensitivity* μ starts at zero and accumulates, at every step, the chain-rule contribution $(\partial\Phi_s/\partial n)^T\lambda$ of that step's Manning dependence. When the sweep reaches $t = 0$, μ is the complete misfit gradient dJ_{mis}/dn and λ is dJ_{mis}/du_0. Both transposed Jacobians are the analytic operators of Sections 3–4 — the assembled RHS Jacobian for the state propagation, the two-nonzero parameter Jacobian for the μ accumulation — applied within `TSAdjoint`'s discrete adjoint of the chosen integrator.

The outer update is TAO's bound-constrained limited-memory quasi-Newton method BLMVM [16] (Algorithm 3). The loop terminates on the standard bound-constrained test: the projected gradient g^P — the gradient with components pointing out of the feasible box zeroed — drops below an absolute tolerance, or an iteration budget is reached. The same loop serves every parameterization in this paper: per-region, per-cell, and land-cover-class fields differ only in the expansion of the parameter vector onto cells (line 3 of Algorithm 1) and the corresponding summation of μ back onto parameters.

Algorithm 1 Strong-constraint 4D-Var calibration of the Manning field, as implemented with TAO/BLMVM and `TSAdjoint`. The window map $M_{t_{k-1}\to t_k}$ is the composition of m steps of the discrete forward stepper (explicit RK, backward Euler, or ARK-IMEX); Algorithm 2 differentiates exactly this stepper, so the gradient is exact to roundoff (and inner-solver tolerance, Section 4) for the computed trajectory.

Require: prior/initial guess n_0, bounds $[n_{\min}, n_{\max}]$, observations y_k at times $t_k = k\,m\,\Delta t$ ($k = 1,\dots,K$), gauge operator H, error σ, Tikhonov weight β, gradient tolerance ε_g, iteration budget $I_{\max}$

1: $n \leftarrow n_0$
2: **repeat**
3: set the model's cellwise drag coefficient from n ▷ identity for per-cell n; constant over each region or land-cover class for the reduced parameterizations
4: $u \leftarrow u_0$ ▷ fixed initial state (strong constraint)
5: **for** $k = 1,\dots,K$ **do** ▷ forward sweep; trajectory checkpointed — every step, or a revolve schedule for long windows (Section 6)
6: $u \leftarrow M_{t_{k-1}\to t_k}(u;\, n)$
7: $r_k \leftarrow Hu - y_k$
8: **end for**
9: $J \leftarrow \dfrac{1}{2\sigma^2}\sum_{k=1}^{K}\|r_k\|^2 + \dfrac{\beta}{2}\|n - n_0\|^2$
10: $(\lambda, \mu) \leftarrow \textsc{AdjointSweep}(\{r_k\})$ ▷ Algorithm 2: $\mu = dJ_{\text{mis}}/dn$
11: $g \leftarrow \mu + \beta\,(n - n_0)$
12: $n \leftarrow \textsc{BlmvmStep}(n, J, g)$ ▷ Algorithm 3
13: **until** $\|g^P\| \le \varepsilon_g$ or $I_{\max}$ iterations
14: **return** n

All experiments use a wet planar dam-break configuration (50 cells, two regions; synthetic noise-free observations; TAO/BLMVM with bounds $n \in [0.01, 0.2]$).

Algorithm 2 AdjointSweep: segmented discrete-adjoint backward sweep (`TSAdjointSolve` with `TSAdjointSetSteps`). Φ_s denotes the forward map of step s, $u_s = \Phi_s(u_{s-1}; n)$; its transposed Jacobians are built from the assembled RHS Jacobian $\partial f/\partial u$ (Section 3) and the parameter Jacobian $\partial f/\partial n$ according to the integrator's discrete-adjoint formula.

Require: checkpointed trajectory $\{u_s\}_{s=0}^{Km}$, residuals $\{r_k\}_{k=1}^{K}$ at steps $s = km$, gauge operator H, error σ

Ensure: $\lambda = dJ_{\mathrm{mis}}/du_0,\ \ \mu = dJ_{\mathrm{mis}}/dn$

$\lambda \leftarrow \sigma^{-2} H^T r_K$ ▷ λ holds $\partial J_{\mathrm{mis}}/\partial u(t_s)$ throughout
$\mu \leftarrow 0$
for $s = Km, \ldots, 1$ **do**
  restore u_{s-1} (and stage values): read the checkpoint, or recompute forward from the nearest one when the checkpoint schedule demands
  $\mu \leftarrow \mu + \left(\partial\Phi_s/\partial n\right)^T \lambda$ ▷ accumulate this step's Manning dependence
  $\lambda \leftarrow \left(\partial\Phi_s/\partial u_{s-1}\right)^T \lambda$ ▷ backward model step (transposed step Jacobian)
  if $s - 1 = km$ for some $1 \le k < K$ **then**
    $\lambda \leftarrow \lambda + \sigma^{-2} H^T r_k$ ▷ observation jump: u_{km} enters J_{mis} directly
  end if
end for
return (λ, μ)

Algorithm 3 BlmvmStep: one iteration of TAO's bound-constrained limited-memory variable-metric method [16]. $P_{[n_{\min}, n_{\max}]}$ is componentwise clipping to the bound box; the curvature pairs $\{(s_i, y_i)\}$ persist across calls. The projected gradient used in the convergence test of Algorithm 1 is $g_i^P = g_i$ for n_i strictly inside the box, $\min(g_i, 0)$ at the lower bound, and $\max(g_i, 0)$ at the upper bound.

Require: iterate n, objective J, gradient g, stored pairs $\{(s_i, y_i)\}$, bounds $[n_{\min}, n_{\max}]$

$d \leftarrow -B\, g^P$ ▷ $B \approx (\nabla^2 J)^{-1}$ by the L-BFGS two-loop recursion over the stored pairs
find $\alpha > 0$ with a projected line search on $n(\alpha) = P_{[n_{\min}, n_{\max}]}(n + \alpha\, d)$ satisfying sufficient decrease of J ▷ each trial α costs one forward-plus-adjoint evaluation (lines 3–11 of Algorithm 1)
$n^+ \leftarrow n(\alpha)$; evaluate g^+ at n^+
$s \leftarrow n^+ - n;\ \ y \leftarrow g^+ - g$; store (s, y) if $s^T y > 0$, discarding the oldest pair beyond the memory limit
return n^+

Observations	β	TAO its	converged	rel. L^2 error
terminal only	10^{-4}	688	yes	39.5%
8 times	10^{-4}	200 (cap)	no	19.9%
20 times	10^{-5}	1864	yes	**8.0%**

Table 2: Per-cell twin recovery vs. observability (planar dam break, 50 gauges, 150 parameters, 40-step window).

Per-region. Truth $n = (0.03, 0.06)$ by region, initial guess 0.02 uniform, terminal-time observations at every second cell. BLMVM recovers both values to 4.1×10^{-7} maximum relative error.

Per-cell with Tikhonov regularization. Truth is a two-zone field split at the domain midline; parameters are per-cell (three per observation in the densest case), with $J \mathrel{+}= \frac{\beta}{2}\|n - n_0\|^2$ about a constant prior. Recovery is assessed over observable cells (wet and moving at the final time; dry cells are untouched by friction and correctly remain at the prior). The controlling factor is observability (Table 2): a single terminal snapshot leaves the problem badly underdetermined (39.5% error at full convergence); eight observation times already halve the error (19.9%, and still descending when stopped at a 200-iteration cap); and 20 observation times over the same window reach 8.0%, the remainder being genuine null space held by the prior. The exact analytic Jacobian reduced the cost of an adjoint sweep by roughly 6× relative to the FD-coloring Jacobian, making the converged experiments interactive (∼2 minutes on one laptop core).

What noise-free twins do and do not establish. The recoveries above are statements about the *machinery* — gradient correctness end to end — and it is important to be precise about why they are not statements about calibratability. On this 40-step (1.8 s) window, the entire Manning-induced observation signal — the RMS difference between the truth run's observations and the prior run's — is about 5 μm of water height (measured from the initial misfit). Noise-free, the optimizer can descend on a micrometer signal indefinitely; with synthetic Gaussian observation noise (a deterministic, decomposition-independent perturbation of the twin observations), 0.1 mm of noise destroys the per-cell recovery ($0.20 \to 1.8$ relative L^2 at the same iteration budget) and 1 mm destroys even the two-parameter per-region recovery ($4 \times 10^{-7} \to 0.83$, the optimizer driving both values to a bound); no Tikhonov weight in a sweep up to $\beta = 1$ rescues it. The lesson is not that the method is fragile — it is that observation noise turns calibratability into a signal-to-noise question, and friction must act long enough to imprint a signal above the noise floor. On the basin-scale configuration of Section 7.1 the parameter-induced peak-WSE signal is ∼0.1 m after one simulated hour — five orders of magnitude above the toy twins' — which is the regime where survey-grade noise (0.1–0.3 m for high-water marks) becomes a fair fight. [TODO: o30: noisy (0.15 m) basin-scale HWM twin — runs queued, numbers to follow.]

6 Implicit stepping, enabled and required

The explicit drag term is stiff in thin water: linearizing gives a stability limit $\Delta t \lesssim 2h^{4/3}/(gn^2|v|)$, which for sheet flow at centimeter depths is seconds — far below the gravity-wave CFL limit of kilometer-scale meshes. Our Hurricane Harvey test configuration (Houston at 1 km, $\Delta t = 30$ s) confirms this empirically: the explicit source treatment produces NaN within the first objective evaluation, in quantitative agreement with the analysis.

The assembled exact Jacobian makes the remedy nearly free: backward Euler via PETSc's theta method, with SNES driven by the same Jacobian, carries the Harvey window at $\Delta t = 30\,\mathrm{s}$, and the theta-method adjoint passes the same FD gates (Table 1).

We also implemented the sharper tool: an ARK-IMEX splitting with the Manning drag as the stiff implicit part (`temporal: arkimex`). The implicit stage system is block-diagonal — an independent 3×3 Newton per cell, no global solve — while fluxes and bed slope remain explicit under the gravity-wave CFL limit. Its adjoint passes the same gates (2.0×10^{-8} for dJ/du_0, 2.5×10^{-6} for dJ/dn) and it carries the Harvey window at $\Delta t = 30\,\mathrm{s}$ at 6.7 s per forward-plus-adjoint sweep, with gradients matching backward Euler to three digits. The splitting also runs under RDycore's libCEED operator backend — the explicit side uses the friction-free source Q-functions prepared in PR #359, and the two backends agree to machine precision (5×10^{-16} relative) after 100 ARK-IMEX steps; a device-resident implicit part is the natural follow-up.

Three implementation facts others will hit:

- PETSc's ARKIMEX adjoint dereferences *both* the implicit- and explicit-part parameter Jacobians, so a zero explicit-part `TSSetRHSJacobianP` must be registered even when the parameter appears only implicitly.
- The embedded step-size controller must be disabled for finite-difference gradient validation: perturbed solves otherwise take different step sequences, which shows up as an apparent $\sim 10^{-4}$ gradient error that vanishes with fixed steps.
- PETSc's in-memory checkpointing trajectory cannot tolerate a *segmented* forward solve (one `TSSolve` per observation window). `TSSolve` sets its converged reason before the final trajectory write of every call, which the memory trajectory interprets as the end of the forward run, corrupting the checkpoint schedule at every interior observation time. The forward sweep must be a single `TSSolve` with observations recorded by a monitor; the segmented backward sweep is unaffected.

Newton robustness is set by residual discontinuities, not by conditioning. [Pre-Wednesday note: the two items below are new findings from the land-cover-prior runs; both fixes are now implemented and gated (the boundary fix as an opt-in transmissive condition, open for correction — open question 2–3). Paper numbers reported elsewhere are unaffected — they come from smooth-n configurations that never trigger either — but this is the honest account of what implicit stepping costs on a real roughness field.] Where implicit stepping fails on this problem, it fails for a reason that is invisible in the smooth twin experiments: the residual is discontinuous, so Newton has nothing to converge *to*. At the land-cover roughness prior ($\bar{n} = 0.10$) rather than the twins' smooth $n \approx 0.03$, backward Euler descends cleanly for a dozen iterations and then freezes, with an $O(10^{-13})$ line-search step changing the residual norm by a factor of eight — the signature of a jump, not of ill-conditioning. Two mechanisms are responsible, both catalogued in Section 8: the drag's wet/dry cutoff, whose amplitude scales as n^2 and therefore only becomes disabling at realistic roughness, and the critical-outflow boundary's stagnation switch. Regularizing the drag velocity removes the first and restores honest convergence over the early transient. The second we removed by replacing the outlet condition itself: a transmissive (zero-gradient) free outflow — the ghost state copies the interior state, the convention of most finite-volume shallow-water codes — is continuous in the interior state by construction, and its ghost map is the identity, so the analytic Jacobian and adjoint carry it exactly (FD-gated like every other block). The effect is not an improvement but a cure: a rain-forced control that Newton could not carry past step 498 under critical outflow (residual pinned at 1.05×10^{-3} relative, tolerance relief buying steps linearly) completes the same window with *zero* failures, 598 of 600 solves converging in two Newton iterations; a one-hour rain-forced calibration window — forward, adjoint, and quasi-Newton line search — runs clean end to end; and a 72-hour rain-forced forward through the Hurricane Harvey crest (259,200 implicit steps on the 2.93M-cell mesh) completes without a single Newton failure. Before the replacement we absorbed the pin by accepting the iterate at a relative nonlinear tolerance of 10^{-3}; that this was tolerable rather than merely expedient is worth stating precisely: a land-cover-class twin calibration under that setting recovers all fifteen class values to machine precision (objective reduced from 1.5×10^7 to 6×10^{-7} in twenty quasi-Newton iterations), so the gradient path was unimpaired by the loosened inner solve. The general lesson for adjoint retrofits is that nonsmoothness in an operational flood code is not only a differentiability question — it is a nonlinear-solver question, and the two discontinuities that mattered here were both invisible until the parameter field was realistic.

Checkpointed adjoints for long windows. A naive stored trajectory for a full 36-hour Hurricane Harvey window at the 30 m mesh's CFL-limited $\Delta t = 0.25$ s (518,400 steps, 2.93M cells) would occupy tens of petabytes. PETSc's checkpointing trajectory (`-ts_trajectory_type memory` with the revolve optimal-schedule library) replaces storage with recomputation. We validated it across the full hierarchy. On the dam-break twin it reproduces the disk-trajectory calibration and gradient gates bit-for-bit for explicit RK, backward Euler, and ARK-IMEX (the implicit integrators need the same tight inner tolerances as in Section 4, since checkpoint recomputation re-solves the stage systems). On the 30 m mesh, a forward-plus-adjoint sweep with only *three* states checkpointed in memory matches the disk-trajectory gradient to all printed digits — and runs slightly faster, because recomputation is cheaper than trajectory I/O at this scale. For the full window, a few hundred checkpoints ($\sim$14 GB aggregate) bound the recompute overhead by a factor of about four, making the long-window adjoint memory-feasible; the remaining cost is the forward compute itself, addressed next.

7 Hurricane Harvey twin experiments: regularization and the real-network limit

The per-cell twin runs on the Houston 1 km Hurricane Harvey mesh (2,746 cells, 4200 s window at $\Delta t = 30$ s). Truth is a two-zone field split at the domain midline, observed at 343 synthetic gauges over 14 observation times, calibrated with implicit stepping and Tikhonov regularization ($\beta = 10^{-5}$). Every cell is wet and moving in this configuration, so all 2,746 parameters are observable. The configuration is deliberately gauge-sparse — 343 gauges for 2,746 parameters, an 8:1 ratio — and it exhibits the classic *semiconvergence* of underdetermined inverse problems (Figure 1). At 300 BLMVM iterations ($\sim$22 minutes on one laptop core, $\sim$4.4 s per forward-plus-adjoint) the recovered map shows the two-zone truth structure cleanly at 26.3% relative L^2 error. Continuing the same weakly regularized optimization to 2000 iterations *degrades* recovery to 38.0%: the misfit keeps decreasing while the parameters drift along observation-null directions, ultimately pinning cells against both bound constraints (Figure 1, center left). Early stopping was acting as the effective regularizer. The behavior disappears when the problem is made well posed: the densely observed dam-break twin of Table 2 converges monotonically to 8%, and a properly regularized Houston run ($\beta = 10^{-4}$, 686 gauges) reaches 19.4% at a 1000-iteration budget with no drift (Figure 1, center right) — the sharpest recovery of the three, with the zone boundary resolved at the truth midline. This is precisely the overfitting risk documented for distributed friction estimation [7], here reproduced in a controlled twin setting. It motivates the regularization studies (total variation, land-cover priors, discrepancy-principle stopping) planned as follow-on work.

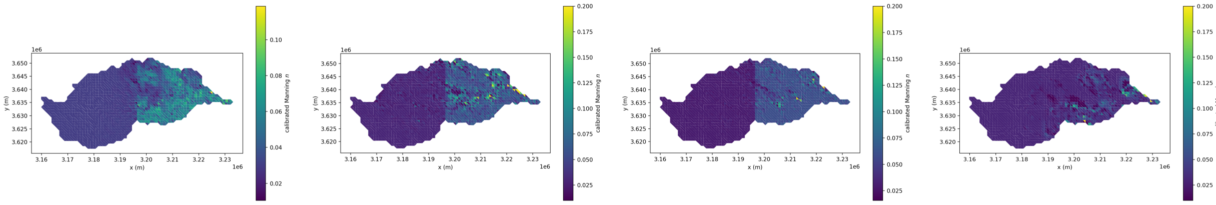

Figure 1: Regularization, semiconvergence, and network density on the Houston twin (2,746 parameters; truth: $n = 0.03$ west, $n = 0.06$ east of the midline; implicit stepping, 14 observation times). Far left: weak regularization ($\beta = 10^{-5}$, 343 gauges) stopped early at 300 iterations — 26.3% error. Center left: the same setup run to 2000 iterations — the misfit is lower but the field drifts along unobserved directions to 38.0% error and saturates the bounds $[0.01, 0.2]$. Center right: proper Tikhonov weight ($\beta = 10^{-4}$, 686 gauges), 1000 iterations — 19.4% error, no drift, sharp zone boundary. Far right: the *real* network — the 17 in-mesh USGS gauges with Harvey stage records, same $\beta = 10^{-4}$, 1000 iterations — the misfit falls 2×10^7-fold while the field recovers to only 44.9%: no trace of the two-zone truth, near-prior almost everywhere with null-space structure pinned against both bounds.

The real gauge network. The synthetic gauge sets above are a controlled ramp; the network that matters is the real one. Twenty USGS gauges fall inside the Houston mesh, seventeen with Hurricane Harvey stage records (HydroShare DOI 10.4211/hs.c037167e497546a1bc1508dfb32a9cff). Re-running the same twin with the observation operator built at those seventeen gauge cells — 238 observations (17 gauges × 14 times) against 2,746 parameters — is a sobering experiment. BLMVM drives the misfit down by a factor of 2×10^7, an essentially perfect fit of the observations, while the recovered field is 44.9% from the truth and saturates both bounds (Figure 1, far right). This is semiconvergence in its purest form: the real network's null space is so large that data fit says almost nothing about parameter recovery, and no Tikhonov weight or stopping rule rescues a per-cell field from 238 observations. The practical conclusion for real-data calibration is to shrink the parameter space to match the network — but, as the next subsection measures, shrinking the parameter space is necessary and not sufficient: whether a reduced parameterization is identifiable depends on the observation geometry and window, not on the ratio of observation to parameter counts.

7.1 Identifiability is a property of the observable: gauges versus high-water marks at 30 m

The natural reduced parameterization for this domain is one Manning value per NLCD land-cover class (15 classes at the 30 m Turning mesh, 2.93M cells), with the published NLCD lookup as prior and truth for twins. Counting is deceptively favourable: even the sparse real network supplies an order of magnitude more observations than parameters. Measurement says otherwise. In a class twin on the 30 m mesh at the real 13-gauge geometry (260 observations, a short early-transient window), the objective falls 128× while the recovered class values remain 66% wrong in relative L^2, worst class 81% — and the failure is structural: classes whose cells are dynamically distant from every gauge (the three forest classes) never move from the starting value, because within the window nothing their roughness does can reach a gauge. The identical configuration observed at 418,076 strided cells recovers all fifteen classes to machine precision, so the failure is the network, not the method. Fifteen parameters, thirteen gauges, and an "overdetermined" observation count:

still unidentifiable.

The same question re-asked with the peak observable comes out differently. Observing the same NLCD-truth twin at the 108 QC-passed high-water-mark cells (peak WSE sampled over a one-hour rain-forced window, implicit stepping, free-outflow outlet; Section 7.2) and calibrating fifteen classes from a uniform start: the objective falls 81×, the peak-WSE error at the marks drops from 0.103 m to 0.017 m MAE, and the area-dominant classes are identified — the three developed-intensity classes, pasture, and crops recover to 0.7–10.5%, together 83% of the domain's cells; area-weighted relative L^2 error is 0.20 (unweighted 0.58). The residual errors concentrate in small-area classes (forests, shrub, emergent wetland, ∼5% of cells), which now *move* — they are sensitive, unlike in the gauge experiment — but overshoot to the bounds along weakly constrained directions: ordinary semiconvergence, treatable with regularization and longer windows rather than a structural absence of signal. Three caveats keep this honest: the one-hour window samples early-transient peaks, not flood peaks (the real calibration windows are 6–12 hours and should only help); the run was capped at 15 quasi-Newton iterations; and twin marks synthesized from the truth trajectory carry no representation error, so this is an identifiability statement — an upper bound on real-data performance, not a forecast of it.

7.2 Changing the observable: high-water marks

Calibration against the actual stage records is blocked by geometry, not software: at all 13 rain-driven USGS gauges in the 30 m domain, modelled water-surface elevation sits 1–11 m *above* observed stage, because a 30 m cell stores one mean ground elevation — essentially the bank of an incised bayou — while the gauge hangs in the channel below it; the offset is time-varying as the real channel fills, so no per-gauge datum correction can repair it. We therefore changed the observable to the USGS high-water-mark archive: surveyed *peak* water-surface elevations on structures and debris lines, i.e. on the floodplain surfaces a 30 m cell does represent, and the benchmark currency of the flood-modeling literature (Inunda reports 0.67 m MAE against Harvey high-water marks [8]). Of 2,364 Harvey marks, 324 fall in the Turning domain; applying to the marks the same QC that disqualified the gauges — is the surveyed elevation above the containing cell's bed? — rejects 62% of them (they are channel surveys), leaving 108 marks of quality fair or better. The misfit compares each mark cell's peak simulated WSE over the window, $\max_k Hu(t_k)$, against the surveyed value; its adjoint needs no new machinery, because each mark's residual can be injected at its own argmax time in the segmented backward sweep of Algorithm 2, which is the exact gradient wherever the peak time is locally constant (Section 8). The parameter gradient of the peak misfit passes the same central-difference gates as every other block (9×10^{-7} against a 10^{-5} gate). The remaining scientific choice — confirmed or corrected by the domain team — is the calibration window, which for marks that record the event maximum must cover the flood peak.

The uncalibrated baseline against the real survey. Because the marks record the event maximum, the honest baseline evaluation must simulate through the crest: a 72-hour rain-forced forward from the available initial state (2017-08-26 18:00, reaching past the Buffalo Bayou crest), NLCD-prior Manning, free outflow, backward Euler at $\Delta t = 1$ s on the 2.93M-cell mesh — 259,200 implicit steps, *zero* Newton failures, two hours on one GPU node. The model wets all 108 marks, and the uncalibrated peak-WSE MAE against the surveyed values is 3.41 m. That number is reported as a starting point, not a result: it bundles roughness error together with initial-condition, rainfall, mesh-elevation, and datum error, and the per-mark decomposition (signed bias, quantiles, and modeled crest times — the last of which also places the calibration window) is the next diagnostic. What the calibration must demonstrate is how much of the 3.41 m a Manning field can close; the literature's calibrated benchmark on this event is 0.67 m MAE [8].

8 Nondifferentiability catalog

Retrofitting an adjoint onto an operational flood code surfaces nonsmooth structure worth documenting:

- **Time step inside the RHS.** Operator-split friction treatments (here: semi-implicit and XQ2018) make f depend on Δt and on intermediate flux divergences; no consistent $\partial f/\partial u$ exists. The remedy is an explicit f plus a genuinely implicit integrator, not a smarter derivative.

- **Entropy-fix branches.** The critical-flow fix is piecewise smooth; branch-faithful one-sided derivatives pass FD checks even on transcritical states (Table 1).

- **Critical-outflow stagnation.** The outflow ghost map $h_g = (q^2/g)^{1/3}$ has an unbounded one-sided derivative at $q = 0$, and the inflow branch switches there: a genuine subgradient point. Unlike the others in this list it is not merely a derivative subtlety — the branch switch is a discontinuity of the *residual* itself, since the $u_\perp < 0$ branch zeroes both states (a wall) while the $u_\perp \geq 0$ branch imposes the critical ghost, so the edge flux jumps by the full wet-onto-dry Roe flux, $O(gh^2/2)$, as a boundary cell crosses stagnation. With an implicit integrator this stalls Newton outright: the line search collapses to $\lambda \sim 10^{-6}$ with the residual pinned at a floor of 10^{-4}–10^{-5} relative (Section 6). Explicit stepping never sees it. The cure is to change the boundary condition, not the solver: a transmissive (zero-gradient) free outflow — ghost state equal to the interior state, the standard free outflow of finite-volume shallow-water codes — is continuous by construction, has an identity ghost map in the Jacobian, and removes the stall entirely (Section 6). We provide it as an opt-in condition alongside the original.

- **Drag at the wet/dry cutoff.** The drag $gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ gated at $h_{\min}$ with a plain $\mathbf{q}/h$ velocity is likewise residual-discontinuous: a cell crossing the cutoff with retained momentum switches an $O(h^{-7/3})$ term on and off. The amplitude scales with n^2, so it is invisible in smooth-n twin experiments and disabling at land-cover roughness. Evaluating the drag with the ANUGA-regularized velocity $\mathbf{u} = \mathbf{q}h/(h^2 + h_{\text{anuga}}^2)$ removes it (the drag then vanishes as $h^{5/3}$), and reduces identically to the plain form at $h_{\text{anuga}} = 0$.

- **Wet/dry branches.** The RHS's dry cutoffs are honored branch-faithfully in every block; fully wet validation cases keep FD comparisons clean, and dry cells are correctly insensitive to their roughness.

- **The argmax in the peak-WSE observable.** The high-water-mark misfit (Section 7.2) depends on the state through $\max_k Hu(t_k)$: piecewise smooth, with kinks where two sampled times tie for a mark's peak. The adjoint injects each mark's residual at its recorded argmax time, which is the exact gradient away from ties (a measure-zero set); but the peak time *moves* across optimizer iterates, so the objective the quasi-Newton method sees is only piecewise smooth and its line search can straddle kinks. Practice is untroubled — the FD gates pass away from ties and the twin calibrations descend monotonically — but this sits on the same practice-ahead-of-theory footing as every max-type observable in calibration, and we say so rather than hide it.

9 Conclusions and outlook

An exact, verified Jacobian turns out to be the single artifact from which everything else follows cheaply: adjoint state and parameter sensitivities, gradient-based calibration, and implicit stepping. The capability is production-ready in the sense that matters for an operational code — behind a flag, FD-gated in CI, and with the verification baseline (FD coloring) kept alive in the configuration system.

Next steps, in order: the Houston per-cell demonstration with real USGS Harvey gauges (datum QC pending); regularization studies (total variation, land-cover priors) for the ill-posedness that our observability results quantify; and coupling the gradient path to PyTorch through the existing PETSc–PyTorch bridge for learned-physics and hybrid data-assimilation experiments.

A closing note on cost. Checkpointing makes the long-window adjoint memory-feasible, but at the 30 m mesh's CFL-limited $\Delta t = 0.25$ s a full 36-hour Hurricane Harvey gradient still costs several forward sweeps over half a million steps. The affordable path stacks reductions that are all now in place. First, fully implicit stepping at CFL-free time steps: backward Euler with the assembled Jacobian carries the 30 m, 2.93M-cell Harvey-region mesh at $\Delta t = 1$ s — four times the explicit limit, with Newton converging in about four iterations per step behind a block-Jacobi-preconditioned Krylov solve; larger implicit steps are under investigation. Second, GPU execution of the entire gradient pipeline: the right-hand side, the assembled Jacobian and its transpose solves, the preconditioner setup, the parameter Jacobian, the observation operator, and the revolve-checkpointed trajectory are all device-resident. On four A100 GPUs, a forward-plus-adjoint gradient of a one-simulated-hour window on that mesh (3,600 implicit steps, 400 in-memory checkpoints) takes about nine minutes, and one TAO iteration of a dense-gauge twin calibration (418,076 gauges, 39-step window) takes 1.7 s — roughly 19× one 64-core CPU node of the same system. Device and host runs produce bitwise-identical trajectories under a common compiler and gradients identical to every printed digit, so the finite-difference gates of Table 1 carry over unchanged. Third, land-cover-class parameterization (tens of TAO evaluations instead of hundreds) and calibration windows placed on the storm peak rather than the full event.

Red/tiger-team review: the weakest parts of this draft

[Temporary section, removed before submission. This is our own adversarial read of the draft — the places a hostile reviewer (or a careful friend) will push, ranked by how much damage they could do. Most map directly onto the numbered questions in the next section; cross-references are given as Qn.]

1. **There is no real-data result in the paper.** The abstract's differentiator — production code made differentiable, versus reduced solvers rewritten in ML frameworks — invites the comparison the paper cannot yet win: Inunda reports 0.67 m MAE against *surveyed* Harvey high-water marks; every calibration here is a twin. The real-gauge experiment we do report (17-gauge network, misfit down 2×10^7-fold, field 45% wrong) is honest and instructive, but it is a negative result about the network, not a demonstration of skill. Until a real-observation number exists, the introduction's framing overdraws. *Mitigation in progress:* the USGS high-water-mark archive is acquired and QC'd (2,364 marks; 324 in-domain; 108 usable after the below-bed and quality filters), a peak-WSE misfit with an exact argmax-time adjoint is implemented and FD-gated, the identifiability twin at the real 108-mark geometry appears in Section 7.1, and the *uncalibrated* real-data baseline is now measured (Section 7.2: 3.41 m MAE over a crest-covering 72-hour forward). Still missing: the calibrated number — which

requires the window-placement decision and, likely, a checkpointed peak-time initial state — and a bias decomposition separating roughness error from IC/rain/mesh error. Until then the paper claims the capability and the baseline, not the closed loop. (Q1, Q2)

2. **The semiconvergence cure is asserted by a counting argument our own evidence undercuts.** Section 7 proposes the land-cover-class parameterization because 238 observations "overdetermine" $\sim$13 classes 18$\times$. But observation *count* is not information content: in an internal class-parameterization twin at the real 13-gauge geometry (short window), the objective fell 128$\times$ while the class values finished 66% wrong — three classes never moved because no water reachable from their cells crosses a gauge in the window. The honest statement is that identifiability depends on the observation *geometry and window*, and must be measured, not counted. *Addressed:* the counting argument is gone; Section 7.1 now reports the measured study — 13 gauges fail (classes distant from every gauge never move), 108 high-water marks identify the classes covering 83% of the domain — with its caveats (short window, iteration cap, no representation error) stated. Remaining exposure: the caveats themselves; a reviewer may ask for the 6–12-hour-window version, which is exactly the planned production run. (Q2)

3. **The verified gradients are not the production gradients.** Every FD gate in Table 1 is passed at tight inner tolerances; the planned production configuration runs the inner Krylov solver at `rtol` 10^{-2}, where the adjoint-vs-FD mismatch is inner-solve-limited at 6×10^{-4}–4×10^{-3}. In every test this does not affect BLMVM convergence. *Partially addressed:* Section 4 now states the production tolerance, the measured gradient error, and the spot-check policy. Still open: a principled footing (inexact-gradient quasi-Newton has known sufficient conditions) rather than an empirical one. (Q3)

4. **"Implicit stepping is required" is conditional on a choice we made.** The production code handles drag stiffness with operator-split friction treatments; they are unusable for the adjoint because they bury Δt inside the RHS, and *that* is why the adjoint path needs an explicit source plus an implicit integrator. The section title claims more than this. Also missing is the honest wall-clock comparison at 30 m: backward Euler at $\Delta t = 1$ s costs $\sim$4–5 Newton iterations per step behind a Krylov solve versus explicit RK at the CFL-limited 0.25 s — the net factor is modest and should be printed, not implied. Larger implicit steps ($\Delta t \geq 5$ s), which would make the case decisive, are currently blocked by a linear-solver failure the draft does not mention. (Q6)

5. **We changed the physics to fix the solver, and the changes await domain sign-off.** The regularized drag velocity ($h_{\text{anuga}} = 0.001$ m, chosen as "smallest that cures the stall"; 0.01 m observed unstable and the mechanism only conjectured) and the transmissive outlet (replacing critical outflow; alters outlet physics; weakly reflective for subcritical flow) are both defensible and both flagged opt-in, but a reviewer can fairly ask whether the calibrated n compensates for boundary and regularization choices rather than physics. Sensitivity of the calibrated field to h_{anuga} and to the outlet choice should be demonstrated, not asserted. (Q4, Q5)

6. **The Hurricane Harvey section is a 2,746-cell twin.** All 30 m-mesh, 2.93M-cell content in the paper is cost measurement; no calibration at 30 m is shown, and a reviewer comparing against the 471M-cell forward-validation claim in the introduction will notice the gap. *Addressed:* the section is retitled, and Section 7.1 now contains two 30 m, 2.93M-cell class calibrations (gauge and high-water-mark twins) run end to end on one GPU node.

7. **All twins are noise-free.** σ enters only as a scale factor; no experiment adds observation noise, so the semiconvergence story (early stopping, Tikhonov weights) is undisturbed by the noise level that in practice sets where semiconvergence turns over. *Partially addressed:* a deterministic noise option now exists and Section 5 quantifies the toy twins' 5 μm signal and its collapse under 0.1–1 mm noise — reframing the toy recoveries as machinery verification and the noise question as a signal-to-noise (window length) question. Still open: the noisy basin-scale HWM twin (queued) and, eventually, noise-consistent regularization (discrepancy-principle stopping). (Q7)

8. **The peak-WSE observable, if adopted, is nonsmooth in a new way.** The argmax-time adjoint is exact wherever the peak time is locally constant, and ties have measure zero — but the peak time *moves* across optimizer iterates, so J is only piecewise smooth and BLMVM's line search can straddle kinks. Practice (and our FD gates away from ties) supports it; theory does not. *Addressed:* the nondifferentiability catalog now carries this entry explicitly; the theoretical footing remains open by construction. (Q2, Q3)

9. **Reproducibility currently rests on an unmerged PETSc fork.** The device-resident assembled-Jacobian path (blocked-COO BAIJ assembly on GPU) exists only on our PETSc branch. Until it is upstreamed or the paper's results are reproducible with stock PETSc (AIJ path, at some cost), the artifact story is weaker than the "behind a flag, FD-gated in CI" framing suggests. (Q8)

10. **The novelty claim needed — and got — a literature sweep.** The original "first E3SM component model with built-in discrete-adjoint parameter estimation" was wrong as stated: MALI routinely inverts for basal friction with an automatic-differentiation adjoint of its Albany velocity solver [14, 13]. The claim is now narrowed to the first *transient* discrete adjoint (differentiated through the time integrator) in an E3SM component, with MALI cited in the abstract and introduction. Remaining check before submission: ocean/MPAS adjoint efforts, and whether any E3SM land-model calibration uses a true adjoint rather than ensemble or surrogate gradients.

Open questions for internal review, in order of importance

[Temporary section, removed before submission. Please reply inline or by email — each question names who we think owns it: the science team (Gautam, Donghui, and colleagues), the numerics team (Matt, Jed), or engineering (Jeff). Where we already acted we flag it — we would much rather be corrected now than after the production runs. Nomenclature: "WSE" is water-surface elevation (bed elevation plus water depth), which is what a stage gauge reports once put on the NAVD88 datum; "cell bed" is the single mean ground elevation our 30 m mesh stores for a cell; "HWM" is a USGS-surveyed high-water mark, the peak WSE reached at a point during the event.]

1. **[Science: Gautam, Donghui] What observable should the real-data calibration stand on? We have pivoted to high-water marks — please veto or confirm.** The stream gauges sit in channels our mesh cannot see: at all 13 rain-driven USGS gauges in the Turning domain, modelled WSE sits *1–11 m above* observed stage, only 1 of 13 gauges has observed water above its own cell's bed on the rising limb, and the offset is time-varying (9.7 m $\rightarrow$ 1.7 m over twelve hours at Whiteoak/Main St, where our cell stays essentially dry) — so no fixed per-gauge datum correction can work. *What we did:* acquired the USGS Harvey high-water-mark archive (2,364 marks; 324 in the Turning domain), applied the same

below-bed QC that disqualified the gauges (62% of in-domain marks fail it — they are channel surveys), keeping 108 marks at quality $\leq$ fair, and implemented a peak-WSE misfit (exact argmax-time adjoint, FD-gated). HWMs record the peak, sit on floodplains and structures the 30 m cell can represent, need no hydrograph timing, and are the benchmark currency of the field (Inunda: 0.67 m MAE). *Questions: is peak WSE at HWMs acceptable as the primary observable for the paper? Should gauge stage be retained as a secondary misfit restricted to times when model and observation are both above the cell bed? And is flood-extent (wet/dry) agreement worth adding as a third?*

2. **[Science + numerics: Gautam, Matt, Jed] What identifiability claim do we make, and which parameterization does the data support?** Our evidence so far: 238 gauge observations fit essentially perfectly while a per-cell field is 45% wrong; a class-parameterized twin at the real 13-gauge geometry (short window) drove the objective down 128× with class values still 66% wrong; the same experiment at the 108 HWM locations is queued. The draft's counting argument ("overdetermined 18×") is indefensible as is. *Questions: do we include the full identifiability study (which classes are observable from which networks and windows) as a paper contribution — we think it is one of the strongest things we have — and is NLCD-class-plus-Tikhonov-to-prior the parameterization you want for the production calibration?*

3. **[Numerics: Jed, Matt] Inexact gradients at production solver tolerances.** At the planned campaign setting (`gmres`, right preconditioning, `ksp_rtol` 10^{-2}, `pbjacobi`) the adjoint gradient differs from FD by 6×10^{-4}–4×10^{-3} relative — inner-solve-limited, not a bug; the same configuration at `rtol` 10^{-3} passes the 10^{-5} gates. BLMVM converges indistinguishably in every test (same J to 5–6 digits). *Questions: is a fixed loose inner tolerance with a spot-check at 10^{-3} an acceptable published methodology, or do you want principled inexactness control (e.g., forcing terms tied to the line-search decrease)? And should the adjoint solves use a tighter tolerance than the forward solves, given the gradient inherits both?*

4. **[Science: Gautam + team] Outflow boundary at zero velocity — we implemented a standard fix; please correct the choice if it is wrong.** Our critical-outflow condition switches abruptly at zero normal velocity: below it the boundary acts as a wall, above it it imposes critical outflow, so the flux jumps by roughly $gh^2/2$ as a boundary cell crosses stagnation, which pins the implicit solver (isolated by controls: a reflecting outlet completes the identical run; 5× stronger friction regularization changes nothing; tolerance relief is a treadmill because the pin level follows the local flow state). We implemented the least clever standard treatment: a *transmissive* (zero-gradient) free outflow — ghost state equal to the interior state, as in ANUGA and Clawpack — continuous by construction, differentiated exactly through host and device Jacobians. It cures the stall: the failing control completes with zero failures and two-iteration Newton solves, and a one-hour rain-forced calibration window runs clean end to end. It is opt-in (`type: free-outflow`); the original remains the default. Caveats: exactly non-reflecting only for supercritical outflow; does not prevent inflow if the interior velocity points inward. *Is a transmissive outlet acceptable for Buffalo Bayou, or do you want a Froude-limited or blended form instead?*

5. **[Science: Gautam + team] Bed-friction regularization — implemented, retroactive sign-off requested.** Our friction term $\tau_b = gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ switched off below the depth cutoff makes the equations jump as a cell crosses it; the jump grows like n^2, invisible in the

twins but disabling at land-cover roughness ($\bar{n} = 0.10$). We now evaluate the friction with the ANUGA-regularized velocity, so drag fades smoothly to zero with depth. Off by default; runs use $h_{\text{anuga}} = 0.001\,\text{m}$, the smallest value that cures the stall; 0.003 m is indistinguishable; 0.01 m is unstable (mechanism conjectured: weaker damping lets an iterate reach $h < 0$). *Is this the right regularization to put in the friction source, and is* 0.001 *m acceptable?*

6. **[Numerics: Jed, Matt] Implicit solver headroom: bigger steps and better preconditioning.** Backward Euler with the assembled Jacobian carries the 2.93M-cell mesh at $\Delta t = 1\,\text{s}$ at ∼4–5 Newton iterations per step with `pbjacobi`-preconditioned GMRES; at $\Delta t = 5\,\text{s}$ the very first Newton *linear* solve diverges (300 GMRES iterations, `pbjacobi`) — a preconditioner failure before nonlinearity even enters. We deliberately do not quote an implicit-vs-explicit wall-clock ratio: forward Euler at $\Delta t = 0.25\,\text{s}$ over the same window, on a configuration differing from the implicit run in the integrator and step size alone, produces a *non-finite* state (every one of the 108 marks reads dry), so 0.25 s is not the explicit limit and any ratio measured against it is meaningless. The advective CFL there is only ≈ 0.04 at $h \sim 2\,\text{m}$, so the binding constraint is not the wave speed; we suspect the explicit friction source on rain-wetted thin cells, which is precisely the stiffness the implicit path exists to absorb. Locating the true explicit limit is in progress. *Questions: what would you try first for* $\Delta t \geq 5\,s$ *— a stronger preconditioner (AMG on a reduced system? block ILU? field-split?), pseudo-transient continuation or other globalization once the linear solve holds, or accepting more Krylov iterations — and is per-step Jacobian reassembly (our current policy) the right default, or should we lag it?*

7. **[Science: Gautam + team] Production-run design: window, observation error, gauge trust.** (a) Which phase of Harvey do we calibrate on? Our observability study favours the rising limb; the peak-agreement result and the HWM observable favour the peak. (b) What observation error σ per observable — USGS quality codes give 0.1–0.3 m for HWMs; what for stage? σ sets the data-vs-prior weight directly. (c) We exclude eight reservoir-influenced sites (Addicks/Barker and Buffalo Bayou downstream) because gate releases are unmodelled — is that exclusion list right? (d) Are $n \in [0.01, 0.2]$ sensible bounds for Houston land cover, and should the NLCD field be the prior throughout (replacing the two-zone twins' constant prior)?

8. **[Numerics: Jed, Matt; engineering: Jeff] Upstreaming the PETSc-side pieces.** The device-resident assembled-Jacobian path relies on our PETSc branch: blocked-COO assembly for BAIJ matrices on GPU (`MatCOOUseBlockIndices`; `baijkokkos`/`baijcuda` classes) plus small TSAdjoint fixes. *Do you want this upstreamed as-is (we think it is generally useful — one index pair per block is the natural COO for FV systems), and does the API need changes before it hardens? Until it lands, should the paper's artifact claim be scoped to the stock-PETSc AIJ path?*

9. **[Engineering: Jeff; numerics: Matt] ARK-IMEX design intent and code placement.** We completed PR #359's `SOURCE_ARK_IMEX` path as: Manning drag *alone* in the IFunction (per-cell block-diagonal IJacobian), bed slope and external sources explicit; your CEED-side friction-free Q-functions are wired in and the two backends agree to machine precision. Does that match the intended split, do you have an ARKIMEX scheme/order preference (we use the PETSc default), and did you envision a device-resident implicit part as the follow-up? Also flag any objections to the new file/config naming (`swe_jacobian_petsc.c`, `numerics.jacobian: none|fd|analytic`, `source.method: explicit|ark_imex`) and to

the closure-based Jacobian preallocation (superset of FV edge adjacency; the COO path already tightens it) before this hardens into a PR.

10. **[Engineering: Jeff] Public API and CI.** The adjoint driver reaches through `private/rdycoreimpl.h` (the LETKF-driver pattern). Which calls deserve promotion to `rdycore.h` (`RDySetObservations`, `RDyAdjointSolve`, `RDyGetSensitivities`?), and are Fortran bindings needed at that point? The new tests need a TSAdjoint-capable PETSc and run in seconds — any CI-matrix concerns (the CEED-only lanes correctly skip `numerics.jacobian` configs)?

11. **[All] Venue and authorship** (already asked): target journal (WRR / JAMES / GMD) and the co-author list.

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